

Telemetry for Safety-Critical Railway Systems

EE 600

Claude E. Shannon | claudio.shannon@calpoly.edu

Edith Clarke | edith.clarke@calpoly.edu



CAL POLY

Dr. Gordon E. Moore

Electrical Engineering Department

California Polytechnic State University

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Abstract

This document demonstrates the CalPolySeniorProject class using anonymized metadata and placeholder content themed around electrical engineering. The example includes figures, a table, and an equation to verify common report features. Replace the text, citations, and placeholders with your real system architecture, analysis, and results.

I. Introduction

Electrical engineering designs succeed or fail on constraints: bandwidth, power, timing, reliability, and implementation complexity. A good report makes those constraints explicit and then shows how each design choice trades one resource for another.

This example uses `biblatex` citations. For instance, Shannon's classic result connects capacity, bandwidth, and signal-to-noise ratio (SNR) [1].

II. Background

In practical wireless links, packet delivery is rarely a step function of distance. Instead, there is often a “transitional region” where delivery smoothly degrades as the receiver moves away from the transmitter [2].

II.1. A Small, On-Theme Equation

A common sanity check is whether the required throughput is even plausible. The (idealized) Shannon capacity is:

$$C = B \log_2(1 + \text{SNR}) , \tag{1}$$

where C is channel capacity (bit/s) and B is bandwidth (Hz).

III. System Overview

Figure 1 is a placeholder for an architecture diagram. Replace the boxed region with a real figure (or keep the box if your reviewers love minimalism).

Placeholder Figure 1: System Architecture Diagram

Replace this box with your real figure (PDF/PNG/JPG).

Figure 1: Example architecture diagram placeholder.

IV. Results

Table I demonstrates a simple results table. Use this pattern for simulation sweeps, lab measurements, or performance comparisons.

Table I: Example performance summary for three configurations.

Config	Reliability	Latency [ms]	Energy [mJ]
Baseline	0.82	140	9.5
Tuned	0.93	110	8.1
Robust	0.97	125	9.0

Figure 2 is a placeholder for a results plot. Replace with your actual plot export (recommended: PDF for vector quality).

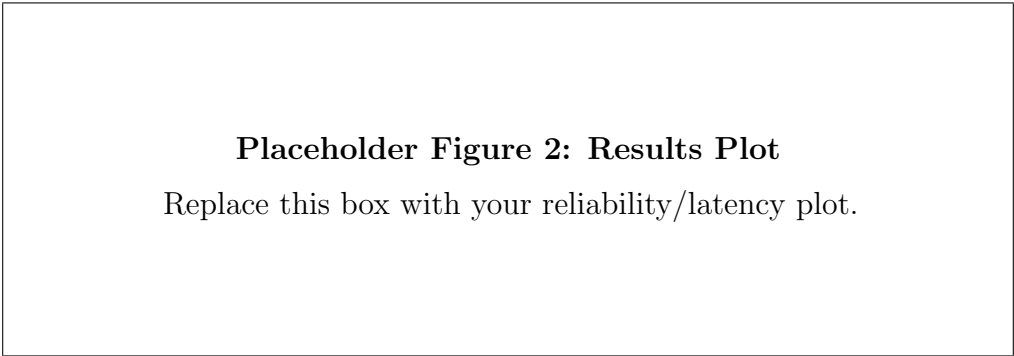


Figure 2: Example results plot placeholder.

V. Conclusion

This example file exists to prove the template is usable for a real report: it compiles cleanly, produces a title page, prints a TOC/LOF/LOT, supports citations, and includes typical technical elements. Replace the placeholders with your system design, assumptions, validation, and final conclusions.

VI. Works Cited

- [1] C. E. Shannon, “A mathematical theory of communication,” *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.
- [2] M. Zuniga and B. Krishnamachari, “Analyzing the transitional region in low power wireless links,” *IEEE SECON*, 2004.

VII. Appendices

Appendix A. Example Appendix

This appendix entry demonstrates the class appendix helper. Add more appendices by calling `\appsubsection{Example}` again.